



CONTRIBUTOR GUIDE

Contributing

Source style, testing, documentation, packaging, and pull-request expectations.

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Use the [Engineering Toolkit](#) for supported trace, replay, camera, and latency workflows from a release archive. Use a complete Git checkout for source changes, tests, documentation generation, deployment, and release packaging.

Thank you for helping improve VirtualGlove. Keep changes focused, readable, and safe for a project that combines camera input, local networking, virtual Linux devices, and a privileged shutdown helper.

User-facing documentation calls the Arduino UNO Q device the **VirtualGlove Controller**. Introduce the hardware relationship once where it helps, then use the product name. Preserve literal compatibility commands, filenames, installed paths, service names, host placeholders, and hardware-specific statements. References to Nintendo's original **Power Glove**, Super Glove Ball's emulated controller, and the separately named [lr-nestopia-powerglove](#) core are factual component names rather than VirtualGlove branding and must remain accurate.

Before changing code

1. Start from the current [dev](#) branch and create a short-lived topic branch.
2. Read [SECURITY.md](#) before changing pairing, tokens, network listeners, downloads, file permissions, uinput, or shutdown behaviour.
3. Check [CONFIGURATION_REFERENCE.md](#) before changing a file format, default, installed path, port, controller mapping, or service unit.
4. Never commit [data/](#), tokens, passwords, pairing codes, model caches and other runtime caches, or locally generated App Lab installation ZIP files.

Report security vulnerabilities through the private process described in [SECURITY.md](#). Do not disclose them in an ordinary pull request or public issue.

Branch and release workflow

[dev](#) is the integration branch for ordinary development. Create feature, fix, and documentation branches from the current [dev](#) branch, then open pull requests back to [dev](#). Do not commit ordinary development directly to [main](#).

[main](#) represents released code. To prepare a release:

1. Confirm the complete quality workflow passes on [dev](#).
2. Finalize the version and move the release notes out of [Unreleased](#).
3. Open and review a pull request from [dev](#) to [main](#).
4. Merge only the reviewed release changes, tag the release, and verify its generated App Lab package.
5. Merge any release-only adjustments on [main](#) back into [dev](#) immediately.

For an urgent released-version fix, branch from [main](#), review and merge the fix into [main](#), publish the corrective release, and then merge [main](#) back into [dev](#). Keep both branches protected against unreviewed or failing changes when the repository host supports branch protection.

Guide layout and audience

Write the installation guide, gameplay guide, camera guide, troubleshooting guide, quick reference, and other user manuals for someone downloading VirtualGlove for the first time. Do not assume that reader has the project's camera, network names, calibration, saved players, or development devices. Lead them from a fresh installation through ordinary use, backups, maintenance, updates, and recovery. Label tested hardware as a recommendation or example rather than presenting it as the reader's configuration.

Keep user guides focused on what people see and what they should do. Put timing, rendering, protocol, benchmark history, and other implementation details in the technical references. Technical guides may describe the project's current UNO Q, supported console, tests, successful experiments, and rejected approaches when that evidence helps another developer reproduce or understand the result. Date or otherwise qualify measurements that may change. Use small contextual images in tables, place related images side by side, and avoid repeating large images when a nearby table already identifies the display. Keep headings, images, and captions together where practical when checking PDFs.

Source style and documentation

Every project-authored executable script, source file, service unit, and application configuration file that supports comments must begin with the standard project header. Preserve a shebang as the first line when one is required. The header must identify:

- project and repository-relative filename;
- concise purpose;
- author and copyright;
- `SPDX-License-Identifier: MIT`;
- a short dated change log;
- `docs/CHANGELOG.md` and Git as the complete history.

Imported or modified third-party source is the exception: retain its original header, authorship, copyright, and license language verbatim. Do not replace or prepend those notices with the VirtualGlove header. Keep project changes in a separate patch and an additive component change ledger, and install or distribute that ledger with the upstream license and notices. If a vendor source tree is accepted later, place it under `third_party/` or `vendor/`; the source audit deliberately exempts those paths from project-header requirements.

Python modules need a useful module docstring. Public classes and functions, plus non-obvious security, protocol, lifecycle, and numerical helpers, need focused docstrings describing their behaviour and safety conditions. Comments should explain decisions and constraints rather than repeat the next line of code.

Use four-space Python indentation, type annotations for interfaces, descriptive names, bounded network and file operations, and existing project patterns. Shell scripts use the shell declared by their shebang; Bash scripts should keep `set -euo pipefail`. Avoid adding a dependency when the standard library or an existing dependency handles the requirement clearly.

JSON does not accept comments. Keep JSON files valid and document their fields, consumer, installed location, defaults, and secret status in `CONFIGURATION_REFERENCE.md`.

Run the source audit before committing:

```
SH
scripts/check-source-docs.py
```

Tests and checks

Run these commands from the repository root on your development computer. The [command reference](#) explains their options.

Core tests must remain independent of a physical camera, VirtualGlove Controller, and console:

```
SH
python3 scripts/run-tests.py --setup
python3 scripts/run-tests.py
python3 -m compileall -q python scripts src tests
```

The first command creates `.venv-test` with the pinned NumPy and headless OpenCV versions used by local image-array tests. Later runs reuse it. The test runner checks those imports before discovery and prints one setup instruction instead of running a dependency-incomplete suite.

Run the documentation and syntax checks:

```
SH
scripts/check-documentation.py
for file in scripts/*.sh; do bash -n "$file"; done
for file in retropie/bin/* retropie/*.sh; do sh -n "$file"; done
```

Add or update tests when behaviour, validation, security boundaries, packet formats, gesture mappings, configuration parsing, or recovery paths change. Do not add tests that only repeat static configuration without protecting a meaningful contract.

Documentation changes

Use two spaces before top-level list markers and keep each item on one source line. The current Help renderer treats wrapped continuation lines as separate paragraphs. Leave blank lines before and after a list, and keep code examples in fenced blocks outside the list. Rendered indentation and spacing are controlled separately by the Help styles and PDF builder; inspect both when publishing a documentation update.

Store project Markdown files under `docs/`, except for `README.md` in the repository root. Write explanatory text in complete sentences and connected paragraphs. Use lists for steps or related items and keep table labels concise. Define unfamiliar terms when they first appear, use consistent names for controls, and check grammar and punctuation before submitting changes. Update all affected guides when a user-visible command, path, control, screen, configuration field, dependency, or troubleshooting procedure changes.

Screenshots, illustrations, and web artwork

Refresh application screenshots from the repository root with:

```
SH
PYTHONPATH=src python scripts/capture-guide-screenshots.py
```

The development environment needs Playwright and Chrome. The script renders the current application templates against isolated sample responses and temporary player state. It never contacts a live Controller or console. Camera areas use a labelled placeholder and pairing inputs use non-secret examples. The capture covers Dashboard, Play, Academy learning and personalization, player settings and restoration, Setup, Games, Help, attract settings, and every guided-pairing state. Shared filenames mean one refresh can affect several guides, so inspect the images before rebuilding the PDFs. The

script also checks the Security network table at phone, tablet, and desktop widths.

Gesture artwork, architecture illustrations, and physical matrix photographs are maintained separately. Preserve the intentional extra-finger artwork. Generate compact Help copies with [scripts/build-help-images.py](#) after editing the originals under [docs/images/gestures](#). Individual web gestures are limited to 320 pixels on their longest side and multi-gesture sheets to 960 pixels; aspect ratio and transparency remain intact. PDF generation continues to use the full-resolution originals.

Website branding icons live under [assets/](#), not in the generated gesture thumbnail directory. The shared shell uses [favicon.ico](#) and the Apple touch icon; PNG variants remain available. Keep the original full logo unchanged. Artwork origins, licensing, and screenshot provenance belong in the consolidated [Third-party notices](#).

Write for the reader's next action

1. Give each guide a clear job: the overview explains the project, installation leads to a working system, reference material defines settings and flags, and game cards help people play. Link between them instead of repeating long explanations.
2. Start a procedure with its goal, prerequisites, and the machine on which it runs. Use numbered steps with direct verbs, then state what success looks like and where to recover from a failure.
3. Explain every command's flags, required values, defaults, and effects in the command reference. Keep copyable examples free of terminal prompts and secrets.
4. Write complete sentences in explanatory paragraphs. Keep labels short, define unfamiliar terms, and remove filler, repeated cautions, and implementation details that do not help the reader act.
5. Give each game card an objective, a gesture-to-control table, and a small first-round exercise. Use light, specific encouragement; keep essential controls easy to find.
6. Check links, images, list numbering, and grammar. Follow each installation from an empty checkout on paper, and distinguish code review, automated checks, and actual fresh-device testing.

These conventions adapt [Microsoft's procedure guidance](#), [Diátaxis's documentation types](#), and [Google's guidance on clear, conversational tone](#). The game-card format also takes inspiration from [Nintendo's beginner game tips](#): introduce the goal, connect a control to its result, and give the player something manageable to try.

Describe changes that affect users in the appropriate category under [Unreleased](#) in [docs/CHANGELOG.md](#). Git history remains the exact implementation record; do not paste a full Git log into individual source headers.

Review and revise the Markdown first. Keep PDF generation separate from the editorial drafting cycle; regenerate the editions when the documentation is approved for publication:

```
SH
python3 scripts/build-docs-pdf.py
scripts/check-documentation.py --require-pdfs
```

When the interface changes, refresh the affected screenshots in [docs/images/](#). The repeatable full capture uses the current page templates, isolated sample data, and an omitted-camera placeholder; it does not contact either device:

```
SH
PYTHONPATH=src python scripts/capture-guide-screenshots.py
```

Install Playwright and Chrome in your development environment before running it. The script also refreshes guided-pairing screenshots and checks the Security network table at phone, tablet, and desktop widths. Capture Dashboard, Play, Glove

Academy practice, Glove Academy tuning, Setup, the Games section, and Help from the running application. Blur the entire camera image before saving a screenshot, keeping instructions and controls readable. Check that no passwords, pairing codes, tokens, or identifying camera details remain. Do not commit unblurred originals. Update image captions and inspect the corresponding PDFs.

Treat Markdown as the documentation source of truth. For publication, commit the approved Markdown and matching regenerated PDFs together. Inspect the affected PDF pages for clipped text, broken tables, missing images, and unintended page breaks before committing.

The public [README.md](#), guides under [docs/](#), and documentation images also drive the Help Center hosted by the VirtualGlove Controller. After a documentation change is merged or otherwise ready to deploy, synchronize and verify that copy with:

```
SH
scripts/deploy-uno-q-wifi.sh arduino@UNO-Q-NAME.local
```

That deployment restarts the VirtualGlove application and checks every Help route, every public PDF, and representative gesture artwork. Confirm the affected page and its **Open PDF** link in a browser after deployment. The cabinet-specific [docs/cheatsheet.md](#) and its quick-reference PDF are intentionally excluded from the public VirtualGlove Controller Help deployment. If the VirtualGlove Controller is unavailable, state clearly that device synchronization and live Help verification remain outstanding.

App Lab installation package changes

Build and verify the model-bundled App Lab installation ZIP with:

```
SH
scripts/build-app-lab-package.sh
scripts/verify-app-lab-package.py
```

The ordinary App Lab installation ZIP must contain production source, configuration examples, documentation, the allowlisted public PDF guides, and the required custom Linux ARM64 MediaPipe wheel used by the Controller, verified Google model, Apache 2.0 license, and third-party notices. It retains the calibration and maintenance support tools outside the engineering inventory in [scripts/package-inventory.py](#), but excludes that engineering inventory as well as private [data/](#), tests, the cabinet quick-reference PDF, caches, and Git history. Development deployments use `--include-engineering`.

Build the optional version-matched research archive with [scripts/build-engineering-tools-package.py](#) `--version VERSION`. The normal release builder creates it alongside the two installers. Do not force-add generated ZIPs to Git. GitHub Actions publishes a short-lived verified ZIP artifact for each successful workflow run; tagged releases may attach a verified ZIP for long-term distribution.

Commits and pull requests

- Use a short imperative commit subject that describes the outcome.
- Explain what changed, why it was needed, and how it was verified.
- Keep unrelated cleanup separate from behavioural changes.
- Call out migration, deployment, compatibility, security, and rollback risks.
- Do not claim hardware verification unless the change was tested on the named device. Automated tests and simulated input should be described accurately.
- Wait for the GitHub Actions quality workflow to pass before merging.
- Target ordinary pull requests at [dev](#); reserve pull requests into [main](#) for reviewed releases and urgent release fixes.

Gesture tuning and firmware validation

Keep Glove Academy and gameplay on the same threshold checks and activation/release states. Preserve the deliberate menu hold and game-specific pulses/toggles. Use **Glove Zap** and **Pull Back** consistently in user-facing labels. Glove Academy and Tune share the scanning-letter matrix renderer and its 160-millisecond frame timing.

Before releasing recognition changes, validate three-step hand setup and individual tuning, including users who skip setup. Exercise V-sign and thumbs-up with different curl ranges, incorrect extended fingers, and incomplete releases. Verify neutral calibration ignores undetected, incomplete-scale, and sub-70% confidence observations; requires 24 accepted frames in production; preserves the previous saved reference until atomic completion; and reproduces a close, not necessarily identical, result for the same simulated stance. Check insufficient samples, tracking loss, overlapping ranges, and calibration changes. Verify preview expiry, save/reload, selected-component reset, rejection of unsupported stored formats without mutation, isolated player settings and progress, stale-tab rejection after player changes or progress resets, and controller suppression throughout tuning. Check that player selection automatically restores that player's saved center, missing centers require centering, and hand-setting imports retain explicit calibration reuse; Start controller remains required. Verify version-4 VirtualGlove round trips and rejection of every older portable format without mutation, invalid calibration rejection, and restart recovery between both restore writes. Automatic suggestions must check separation and a simultaneous full-pose match in at least 90% of accepted samples in each phase, using candidate values and existing personal extension thresholds. Test each extended finger, exact threshold boundaries, default/personal ranges, tolerated noise, and failures occurring in different frames. Failed analysis must clear previews without changing saved settings. Extended-only samples still cannot learn a curled boundary. Manual numerical editing and live-camera validation remain separate concerns.

Perform live camera checks of both menu poses, ordinary finger controls, Glove Zap, Pull Back, and actual game input before release. Successful automated tests, compilation, firmware upload, and bridge calls do not establish real-world recognition quality or visually confirm the physical matrix animation.

Use the complete pinned Zephyr 1.0.0 configuration for compile-only validation, then explicitly upload through App Lab. See the installation guide's [firmware workflow](#). Regenerate all PDF editions after documentation updates. Keep reference screenshots aligned with the current labels and layout. Use the isolated capture workflow, or blur live camera imagery before taking a screenshot.

Two-machine installer releases

Use [Build and publish installation assets](#) to package an explicit release tag. Both installers share `setup-machine.py` and `install-package.py`; changes to host integration must cover first installation, repeat updates, private-settings preservation, and read-only checks. Run `test_install_packages.py` and `test_setup_machine.py`, plus the full test suite. Keep development builds as prereleases. Workflow artifacts are not published release downloads. Physical fresh-device and upgrade tests, cold boot, pairing, and gameplay remain release gates; simulated filesystem tests do not replace them.

Review parking lot

The Changelog records completed reviews and release evidence. Keep unresolved measurement-dependent movement work here rather than maintaining a second release history. Separate camera-to-display validation from routine UI and persistence fixes.